

Market-Neutral ML Strategy on Hyperliquid Perpetuals

Preliminary Backtest Report

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1 Objective and Goal

The aim of this project is to develop, build, and validate a systematic trading strategy that can ultimately be applied to a vault on Hyperliquid. The strategy exploits signals generated by machine-learning algorithms on perpetual futures and converts those signals into long and short trading decisions.

The core philosophy is market-neutral and quantitatively driven. Rather than taking a discretionary view on whether the overall market should rise or fall, the strategy embraces an agnostic viewpoint of the market. Trading decisions are generated from numerical signals, rankings, and predefined portfolio-construction rules. In this sense, the objective is not to forecast the absolute direction of crypto markets, but to extract relative alpha from the cross-section of perpetual contracts.

At this stage, the backtest should be interpreted as a baseline research artifact. It is designed to answer a simple question: *does the raw signal contain enough information to produce a profitable long/short portfolio before adding take-profit logic, stop-loss logic, dynamic sizing, volatility targeting, or other risk-management adjustments?*

2 Current Strategy Setup

The current backtest implements a vanilla long/short portfolio on Hyperliquid perpetuals. Each trading day, the system ranks the eligible perpetual universe using the CrowdCent prediction signal and opens positions according to the top and bottom ranked names.

Table 1: Main backtest configuration.

Parameter	Value
Signal source	CrowdCent ranks
Prediction field	pred_30d
Symbol field	id
Date field	release_date
Portfolio construction	1 long and 1 short per signal date
Trade notional	15 USDC per order
Initial equity	1,000 USDC
Leverage	1.0x
Margin mode	Cross margin
Holding period	30 days per daily-signal quantity
Entry time	19:00 UTC using the hourly open price
Fee rate	0.045% per order
Funding	Applied hourly using funding rates and oracle prices
End-of-backtest handling	Remaining positions are closed at the end

The resulting portfolio is market-neutral at the construction level because it opens both a long leg and a short leg. However, because positions evolve with market prices and are held over time, realized exposure can drift. This report therefore treats the setup as an initial market-neutral baseline rather than a fully optimized portfolio with active exposure control.

2.1 Vanilla Portfolio Assumption

The portfolio tested here deliberately excludes several layers that would normally be considered before deploying capital in production. In particular, the current version does **not** include:

- take-profit rules;
- stop-loss rules;
- dynamic position sizing;
- volatility targeting;
- funding-aware trade rejection;
- drawdown-based de-risking;
- portfolio-level exposure caps beyond the fixed long/short construction;
- adaptive risk-management adjustments.

Therefore, the results should be read as the performance of the raw signal combined with a simple mechanical execution and holding-period policy.

3 Backtest Methodology

The backtester loads CrowdCent ranks, normalizes the symbol names to the Hyperliquid tradable universe, and filters the predictions to the available market data. For each valid signal date, it selects the highest ranked contract for the long book and the lowest ranked contract for the short book. Each order uses a fixed 15 USDC notional and 1.0x leverage.

Positions are monitored hourly. The account equity includes realized profit and loss, unrealized profit and loss, fees, and funding. Funding is accumulated through time and has a direct impact on portfolio equity. Liquidations are approximated using the hourly OHLC data, with the close used as a mark-price proxy and high/low candles used as intrahour liquidation proxies. This makes the simulation useful for research, but it should not be interpreted as an exact replica of the Hyperliquid matching engine.

The test period starts on 2025-06-05. The last filtered signal date is 2026-05-07, and the effective market cutoff is extended by the 30-day holding-period logic. In the available data, the final recorded backtest timestamp is 2026-05-15 10:00 UTC.

4 Backtest Results

4.1 Summary Metrics

Table 2: Main performance metrics.

Metric	Value
Initial equity	1,000.00 USDC
Final equity	1,380.91 USDC
Total return	38.0%
CAGR	41.0%
Maximum drawdown	-14.0%
Annualized hourly Sharpe	1.48
Annualized hourly Sortino	1.41
Calmar ratio	2.96
Maximum gross leverage on equity	1.17x
Total realized PnL	557.43 USDC
Long realized PnL	-1.39 USDC
Short realized PnL	558.82 USDC
Cumulative funding PnL	-167.69 USDC
Total fees	8.83 USDC
Closed-position win rate	61.0%
Long closed-position win rate	48.0%
Short closed-position win rate	73.0%
Profit factor	1.86
Average closed winner	3.01 USDC
Average closed loser	-2.51 USDC
Expectancy per closed trade	0.85 USDC

The backtest ends with 1,380.91 USDC from an initial 1,000 USDC, corresponding to a total return of approximately 38.0%. The realized PnL is strongly asymmetric between the two legs: the short book generated 558.82 USDC, while the long book was approximately flat to slightly negative at -1.39 USDC. This indicates that, in this preliminary run, most of the alpha comes from the short side of the signal.

Funding was a meaningful drag on the strategy. The cumulative funding PnL is -167.69 USDC, which offsets a material part of the realized trading PnL. This is important because a raw signal that looks profitable before funding may become much less attractive once the cost of maintaining perpetual positions is included.

4.2 Equity Curve

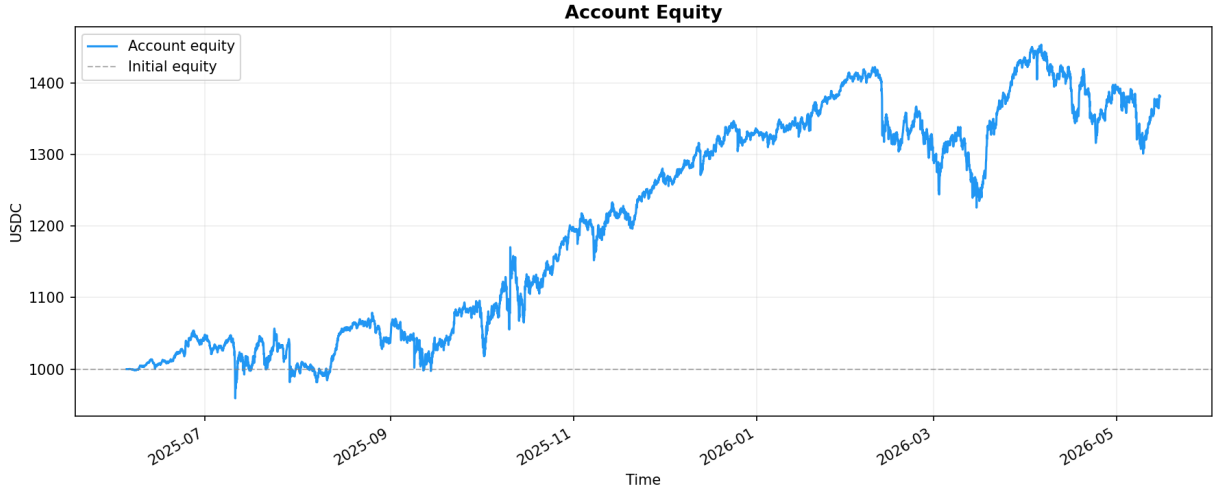


Figure 1: Account equity through time. The dashed line marks the initial 1,000 USDC equity level.

The equity curve shows a positive upward trend over the full sample. Performance is not linear: the account experiences periods of stagnation and drawdown, especially around March 2026 and again near the end of the sample. Nevertheless, the final equity remains substantially above the initial capital.

4.3 Long/Short PnL Decomposition

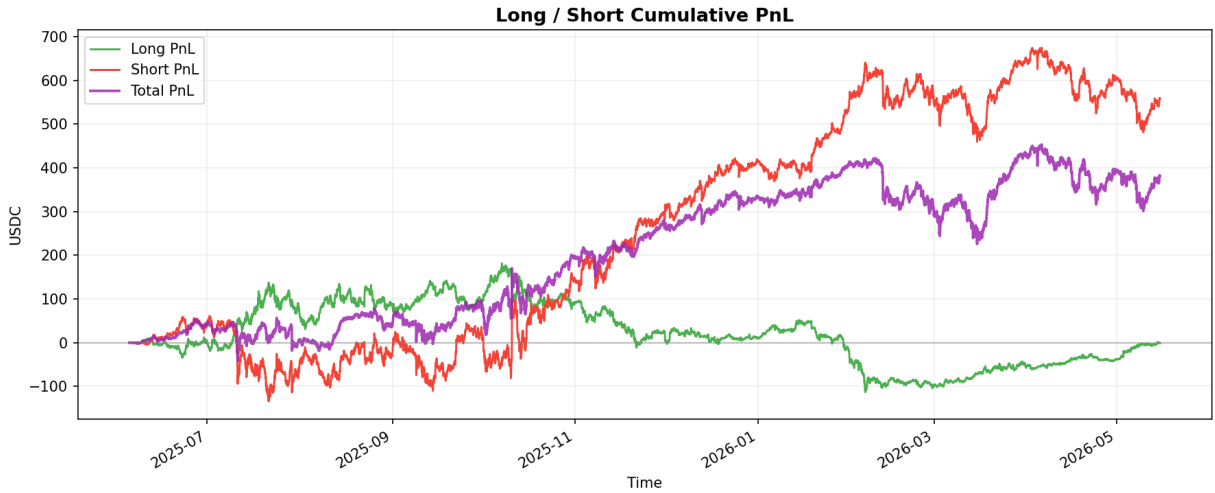


Figure 2: Cumulative PnL split between long and short books.

The PnL decomposition is one of the most informative outputs of the backtest. The short book is the main contributor to performance, while the long book becomes negative for a significant part of the sample. The total PnL therefore largely follows the short-side profitability. This suggests that the current CrowdCent signal is more effective at identifying weak assets than strong assets in this specific configuration and time period.

4.4 Drawdown

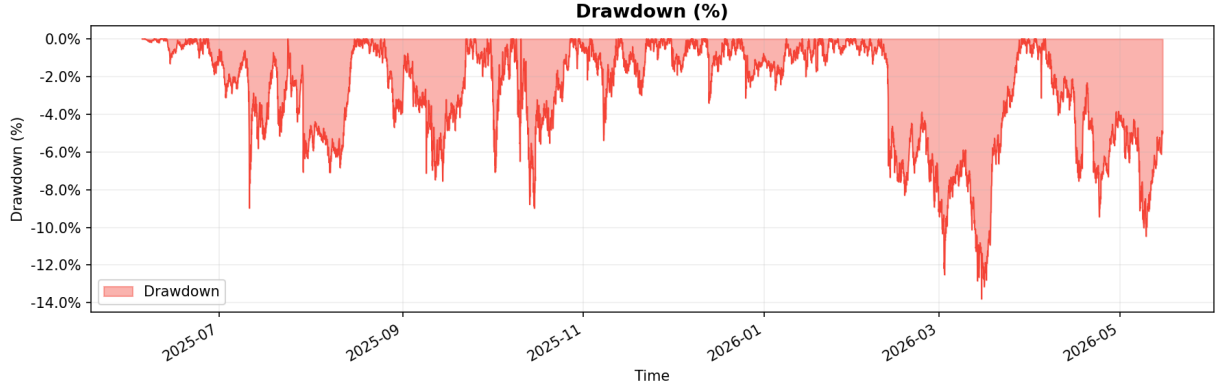


Figure 3: Portfolio drawdown relative to the running maximum of account equity.

The maximum drawdown is approximately -14.0%. The deepest drawdown occurs around March 2026, when the equity curve gives back a meaningful portion of previous gains. Since the portfolio currently has no stop-loss, take-profit, or drawdown-control layer, this drawdown is a direct expression of the vanilla signal-plus-holding-period construction.

4.5 Per-Trade Return Distribution

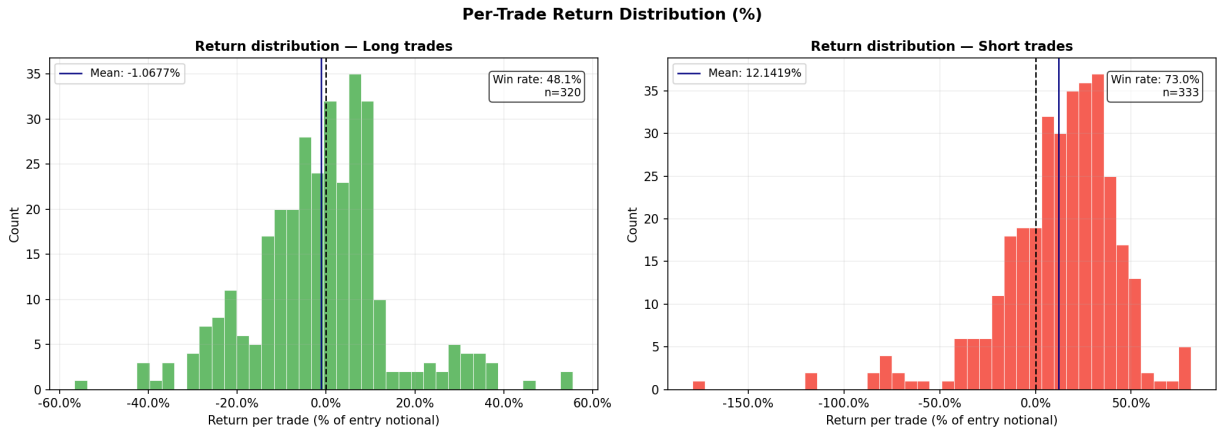


Figure 4: Per-trade return distribution for long and short closed trades.

The distribution confirms the asymmetry visible in the cumulative PnL. The long trades have a win rate of approximately 48.1% and a negative mean return, while the short trades have a win rate of approximately 73.0% and a strongly positive mean return. The short side also contains large negative outliers, which is expected for short exposure because losses can become large when the underlying asset rallies significantly.

4.6 Funding PnL

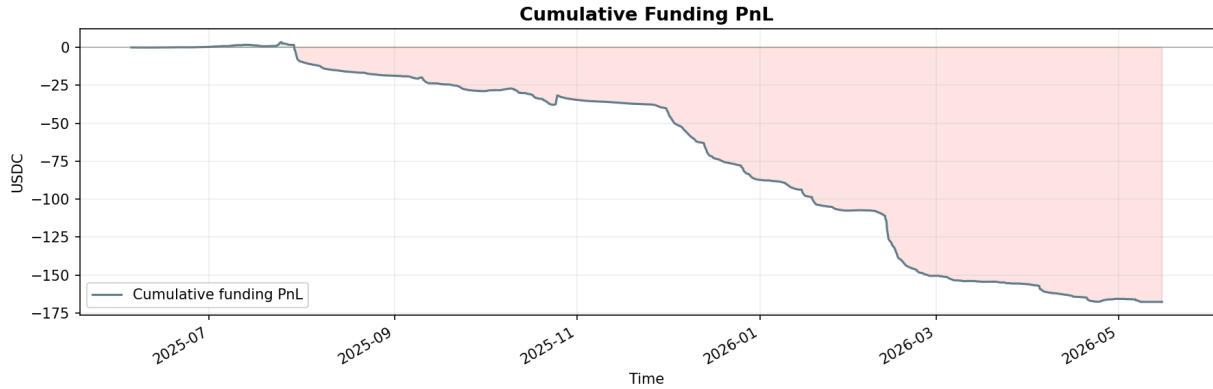


Figure 5: Cumulative funding PnL over the backtest.

Funding is persistently negative over the test. The final cumulative funding PnL is -167.69 USDC. In practical terms, this means the strategy paid a meaningful amount to maintain its perpetual positions. A production version should therefore consider a funding-aware filter or a position-sizing rule that reduces exposure when the expected alpha does not compensate for expected funding costs.

5 Interpretation

The preliminary result is promising because the unoptimized vanilla portfolio is profitable after fees and funding. The final equity is higher than the initial equity, the Sharpe ratio is positive, the profit factor is above 1, and there are no liquidations in this run. This suggests that the signal contains useful information for constructing a long/short perpetual portfolio.

At the same time, the result should not be overinterpreted. The strategy is currently very simple and most of the performance comes from the short side. A robust production system should not rely on a single source of edge without testing whether the behavior is stable across different periods, parameter choices, and market regimes.

The most important research observations are:

- The raw CrowdCent `pred_30d` signal generated positive net performance in this configuration.
- Short trades were much stronger than long trades, both in realized PnL and win rate.
- Funding costs were large and should be explicitly modeled in the next iteration.
- The maximum drawdown of roughly 14% suggests that risk control is mandatory before deployment.

6 Next Steps

The next phase should transform the current vanilla backtest into a risk-aware strategy research framework. The following additions are natural priorities:

1. **Take-profit and stop-loss logic.** Test whether exits based on realized return, trailing stops, or adverse movement reduce drawdowns without destroying alpha.
2. **Funding-aware trade selection.** Penalize or exclude trades where expected funding costs are too large relative to expected signal strength.
3. **Dynamic position sizing.** Scale notional by signal confidence, volatility, liquidity, or recent drawdown.
4. **Exposure control.** Track gross exposure, net exposure, sector/token concentration, and liquidation distance.
5. **Parameter sweeps.** Compare different numbers of longs and shorts, holding periods, leverage levels, and trade notionals.
6. **Vault-readiness checks.** Add execution constraints, slippage assumptions, capacity limits, and operational risk monitoring.

7 Conclusion

This report documents the first complete baseline of a machine-learning-driven long/short strategy on Hyperliquid perpetuals. The current implementation is intentionally simple: it uses CrowdCent signals, fixed notional sizing, symmetric long/short construction, 1.0x leverage, hourly funding, and a 30-day holding rule. It does not yet include take-profit, stop-loss, or advanced risk-management overlays.

The preliminary backtest is positive, ending with 1,380.91 USDC from 1,000 USDC and producing a 38.0% total return over approximately 344 days. The result is mainly driven by the short book, while funding represents a substantial cost. These observations provide a clear direction for the next research iteration: preserve the signal edge, especially on the short side, while improving the risk layer and reducing funding drag.